

2024 M3 Math Modeling Challenge

A Tale of Two Crises:

The Housing Shortage and Homelessness

Executive Summary

To the Secretary of the U.S. Department of Housing and Urban Development,

For the past several decades, shortages in available housing have made finding affordable living units more precarious. These shortages have made the cost of housing grow faster than incomes, making more individuals susceptible to homelessness and its negative effects, such as mental health issues and addiction.

Combating the increasing housing costs and reducing the unhoused population presents a unique challenge in terms of cost, land availability, and population constraints. Our team strives to predict the future housing supply, calculate the changes in the unhoused population, and develop a robust long-term plan to alleviate homelessness that accounts for unexpected stresses such as natural disasters and economic recession. Our two regions of choice are: Seattle, Washington (WA) and Albuquerque, New Mexico (NM).

We first predicted the future supply of housing in Seattle and Albuquerque by using an Auto-Regressive Integrated Moving Average (ARIMA) model, a time series regression and forecasting model that uses auto-correlation to give accurate forecasts from volatile short-term data. The model was trained on a dataset of past vacant housing unit data for the cities of Seattle and Albuquerque. The model ultimately predicted that there will be 19,294 vacant, purchasable housing units in 2034, 24,447 vacant housing units in 2044, and 24,153 vacant housing units in 2074 for the city of Seattle. For the city of Albuquerque, our model predicted an vacant housing unit quantity of 16,731 in 2034, 16,900 in 2044, and 18,468 in 2074.

Next, we used a sequential neural network to determine the homeless population in both of our target regions. By using historical data on vacant housing units and unemployment rates and forecasting future vacant housing units and unemployment rates using an ARIMA model and another sequential neural network, we trained our models to 17.7% and 9.86% error in assessing the Seattle and Albuquerque homeless populations, respectively.

Finally, we developed a distribution-approximation model to predict if a city government's unemployment benefits are sufficient for the homeless population to find affordable housing units. Specifically, we utilized a simple random sample of monthly rent rates (using Seattle as a case study) to determine the monthly unemployment benefits required to achieve the monthly rent rate associated with the homelessness rate.

Overall, this research paper aims to provide a comprehensive analysis of the housing supply and homelessness, with an emphasis on developing long-term solutions so underserved communities can thrive into the future.

We hope these predictive models and solutions can be helpful for policymakers in designing initiatives to combat the affordable housing crisis among their constituencies.

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Question 1: It Was the Best of Times

Restatement of the Problem

In this problem, we are tasked with creating a model to predict changes in the housing supply market of Seattle, WA, and Albuquerque, NM.

Definitions

To accurately convey information about housing, we must first establish some definitions:

Housing supply refers to the number of housing units available on the housing market. This definition encompasses permanent and rented homes.

A **lodging unit** is defined as a short-term, non-permanent housing unit such as a hotel suite, hostel unit, Airbnb, or bed-and-breakfast. These will not be considered within the housing supply.

A **long-term housing unit** is a permanent or rented housing unit that is not part of the housing supply.

A **vacant housing unit** is defined as a permanent or rented housing unit that does not contain one household at the moment.

Assumptions

1. *There will be no significant changes regarding housing policy or zoning laws in the next 50 years.* We are unable to predict the actions of national, state, or local governments because they do not follow a definitive numerical trend. Thus, we will assume that the current stance on housing policy will remain the same.
2. *There will be no major technological advancements in home construction in the next 50 years.* While technological innovations have the potential to drastically reduce the production costs of housing construction, the development and implementation of such innovations can be unpredictable and time-consuming.
3. *Each long-term housing unit will contain one household.* Due to the unverifiable nature of home sharing, we will assume that each household occupies one housing unit (apartment, condominium, single-family unit, townhouse).
4. *Lodging units will not be considered in the housing supply.* Lodging units are not permanent dwellings and the availability of lodging units is affected by different factors than permanent and rented homes, meaning they will not be properly accounted for by our model.
5. *Owners of vacant housing units are willing to sell those properties.* Because vacant housing units include vacancies that are unavailable in the market, we have chosen

to ignore the prevalence of these circumstances because vacant off-market homes are virtually untraceable amongst occupied off-market homes.

6. *Future trends in unemployment rates, housing prices, and population will follow past trends.* Our models extrapolate future behavior based on past behavior and therefore require the continuation of previously exhibited behaviors to make robust predictions.

Variables

Symbol	Definition	Unit
$\hat{V}_x(t)$	Predicted housing supply t years after 2010 for city x	Housing units
$V_x(t)$	Actual housing supply t years after 2010 for city x	Housing units

Table 1: Variable List for Question 1

Model Development

We initially planned to use an array of factors to predict changes in housing vacancy using a Random-Forest regression model, however, the sparseness of data caused this model to become prone to overfitting, which occurs when the model is too closely aligned to a set of data points, which can harm long-term prediction. We also attempted to use Vector Auto-Regression (VAR), however, due to the non-stationary nature of recent data as far back as 2000, the prerequisite conditions for auto-regression could not be met, restricting our analysis to only recent housing vacancy data, which was provided by the Census Bureau (Bureau U.C.S, 2023).

We decided to use an Auto-Regressive Integrated Moving Average (ARIMA) model to study recent trends in housing vacancy. This model uses auto-correlation and moving averages to forecast future trends with greater accuracy. These methods analyze correlations of nearby data points and averages of nearby data points respectively, which, in comparison to a standard regression model, allows ARIMA model to observe extrapolations of more volatile short-term data. Recent data for housing vacancy indeed exhibits volatile behavior, making it difficult to predict with a VAR or Random-Forest model.

We utilized a Python library to perform our ARIMA analysis. First, we verified the stationarity of the short-term housing vacancy data set using an Augmented Dickey-Fuller test. It is crucial that stationary time-series variables, or variables whose statistical properties remain independent from seasonal changes and time, were chosen to easily identify patterns within the data. Using auto-correlation and partial auto-correlation plots, we determined that the

optimal model uses second-order AR terms (two auto-regressive terms) and first-order MA (moving averages) terms. We then fit the model and plotted the results.

We then used the model to predict values for housing vacancy in Seattle, Washington, and Albuquerque, New Mexico, 10, 20, and 50 years from the present day (2024). Finally, confidence intervals were graphically evaluated for our ARIMA plots.

Results

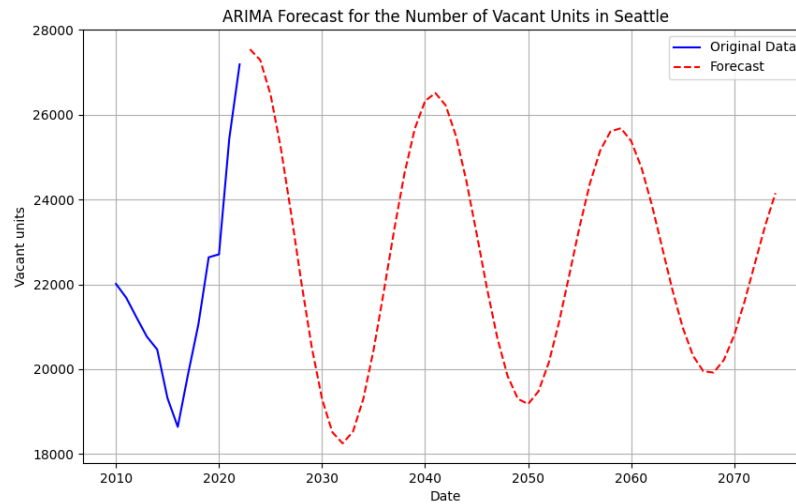


Figure 1: Forecasted values for Housing supply in Seattle

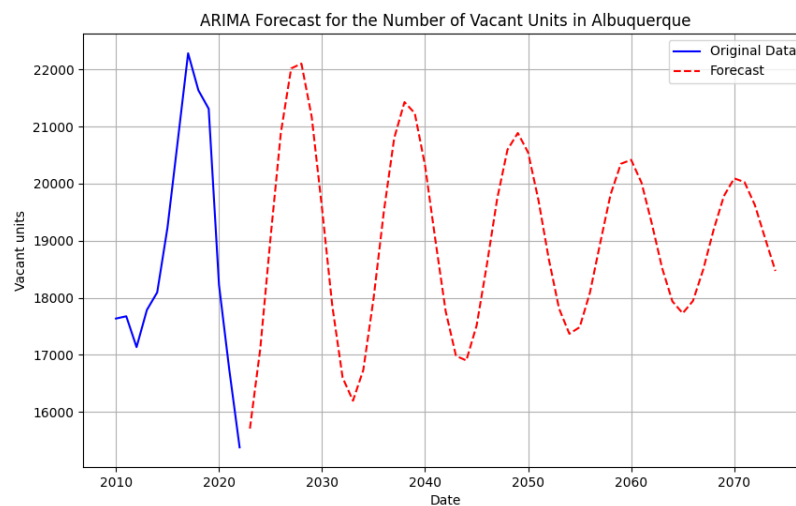


Figure 2: Forecasted values for Housing supply in Albuquerque

Value	Confidence Interval	p-value
Constant	$(1.97 \cdot 10^4, 2.55 \cdot 10^4)$	$< .01$
auto-regression term 1	$(1.637, 2.067)$	$< .01$
auto-regression term 2	$(-1.053, -0.895)$	$< .01$
moving average term	$(-1.801, -0.090)$	0.030

Table 2: ARIMA Results List for Question 1

Our ARIMA model predicted that there would be 19,294 vacant, purchasable homes in 2034, 24,447 vacant homes in 2044, and 24,153 vacant homes in 2074 for the city of Seattle. For the city of Albuquerque, our model predicted a vacant home quantity of 16,731 in 2034, 16,900 in 2044, and 18,468 in 2074. The model satisfies 95% confidence intervals for its constant, AR, and MA terms.

Both applications of our ARIMA model predicted a continuously fluctuating quantity of vacant housing units beyond the historic data. This is likely due to the drastically variable nature of vacancy data, which is reflected in the sinusoid-like behavior that is exhibited in both cases. Notably, however, this volatility is predicted to slowly subside over time, neutralizing to a nonzero level close to the historic average. Based on this observation, our ARIMA suggests that housing vacancy will remain a characteristic of the housing markets in both Seattle and Albuquerque into the foreseeable future if current trends are maintained.

Sensitivity Analysis

To investigate the sensitivity of our ARIMA model, we increased historic values for housing vacancy by 1%. This change increased our predicted housing vacancy quantity from 18,468 to 18,707, a 1.3% increase.

Strengths and Weaknesses

The auto-regressive integrated moving average model allows us to utilize small datasets to generate accurate predictions that suit our situation regarding the vacant housing unit data we obtained. ARIMA also responded well to the volatility of recent data on housing vacancy, being able to build future trends based on both short-term and long-term behavior of our data sets.

However, one weakness of an ARIMA model is its unreliability with long-term predictions. While our 10-year and 20-year predictions about future housing supply are strong, there may be accumulated error regarding our longer-term 50-year prediction, which required an extrapolation far beyond the depth of the available data for housing vacancy.

Additionally, the ARIMA technique does not take into account any other external variables and relies only on previous data points within our chosen data set, which affects the comprehensiveness of our predictions when considering other related factors.

Question 2: It Was the Worst of Times

Restatement of the Problem

In this problem, we are tasked with determining the change in the unhoused population in the next 10, 20, and 50 years for Seattle, WA and Albuquerque, NM.

Definitions

In order to accurately calculate the future unhoused population in our target regions, we must first establish the following definitions:

A person is **unhoused** if they do not have daily access to a permanent or rented housing unit. A person living in a lodging unit will be considered unhoused because lodging units are inherently un-permanent dwellings and do not contribute to the housing supply.

According to the Bureau of Labor Statistics, a person is classified as **unemployed** if they do not have a job, have actively looked for work in the prior 4 weeks, and are currently available for work (Bureau of Labor Statistics, n.d.).

Assumptions

1. *There will be no significant changes in policy regarding unhoused populations in the next 50 years.* Once again, we are unable to predict the actions of governments because they do not follow a definitive numerical trend.
2. *There will be no major technological advancements in home construction in the next 50 years.* While technological innovations have the potential to drastically reduce the production costs of housing construction yielding a significant decrease in the homeless population, there is no accurate way to predict future innovations.
3. *Future trends in unemployment rates, housing prices, and population will follow past trends.* Our models extrapolate future behavior based on past behavior and therefore require the continuation of previously exhibited behaviors to make robust predictions.
4. *There will be no humanitarian crisis happening in the next 50 years.* While events like COVID-19 can significantly affect unhoused populations, we are unable to predict whether they are going to happen or not.

Variables

Symbol	Definition	Unit
$\hat{H}_x(t)$	The predicted number of unhoused people t years after 2009 for a city x	People
$H_x(t)$	Number of unhoused people t years after 2009 for a city x	People
$V_x(t)$	Number of vacant housing units t years after 2009 for a city x	Housing units
$U_x(t)$	Unemployment rate t years after 2009 for a city x	Percentage (%)

Table 3: Variable List for Question 2

Model Development

To predict the number of homeless people H in a given city x , t years after 2009, we created a sequential neural network composed of two variables, namely the number of vacant housing units and the unemployment rate in the respective city. Originally, we had considered four factors: the number of vacant housing units, the unemployment rate, the drug use rate, and the mental illness rate. However, we discovered that gentrification resulting in a lack of affordable housing was a more significant contributor to per capita homelessness than drug use and mental illness in West Coast cities (Warth, 2022). Additionally, we chose to include the unemployment rate in our model because, according to an article published in the Journal of Housing Economics, the unemployment rate was heavily correlated to the homeless population (Corinth, 2017). For this problem, a sequential neural network allows us to assess the impact of these factors to forecast future trends, which is the main caveat to many other forecasting models.

For reference, a neural network is a supervised machine learning algorithm, which uses existing data to forecast future data points. To achieve this, a network consists of an input layer (where the features or independent variables are inputted), hidden layers (which strengthen the accuracy of the neural network), and an output layer (where the predicted value or label is outputted). Each layer consists of several nodes, which are interconnected with the nodes of the next layer using branches. Each non-input node contains a bias and an activation function that are determined by the training data to minimize the error between the predicted and the actual value. Once a model has been trained, new data can be inputted through the neural network to obtain a predicted label.

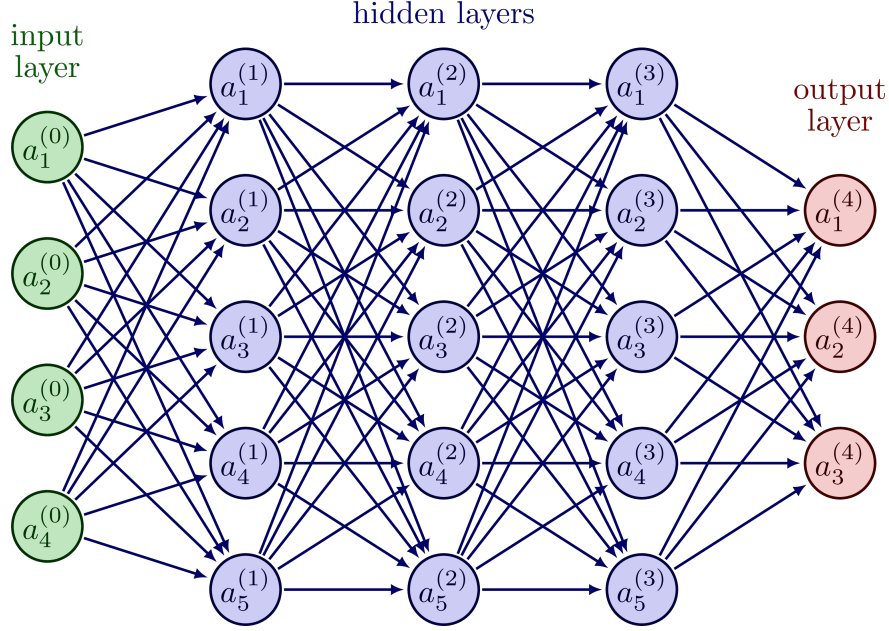


Figure 3: Illustrative diagram of a neural network with one input layer, three hidden layers, and one output layer (Neutelings, n.d.)

Our neural networks consisted of two input nodes for our two features: the number of vacant housing units and the unemployment rate. Using vacant housing and homeless population data provided by the US Census Bureau and sourcing unemployment rate data from the Federal Reserve of Economic Data (FRED), we trained a neural network using the Tensorflow.Keras Sequential Python library. Because the neural network is non-recursive, we determined future vacant housing and unemployment rate data points using our ARIMA model from Q1 and another sequential neural network, respectively (Bureau, U.S.C, n.d.), (Unemployment rate, 2024). Finally, we used these forecasted data points to determine the homeless population in 2034, 2044, and 2074.

Results

Using a sequential neural network on the inflation rate in Seattle depending on the year, it was evaluated that the unemployment rate in Seattle would be 4.988%, 4.874%, and 4.530% in the years 2034, 2044, and 2074, respectively.

Using these values, we achieved the following predictions for the homeless populations in Seattle and Albuquerque:

In Seattle, it appeared that, even with an increase in the number of vacant housing units and a decrease in the unemployment rate over time, the homeless population increased. This value raises many questions; however, according to a Seattle Times article, the increasing cost of housing grows faster than the number of vacant housing units, crowding out many consumers from being able to purchase affordable housing units (Patrick, 2023). Even with the increases in vacant housing units, many low-income individuals are deterred by housing affordability.

On the other hand, in Albuquerque, an increase in vacant housing units and a decrease in the unemployment rate contributed to a decrease in the homeless population. These results should be viewed with caution as the housing population appears to be on the rise. We believe the decreasing predictions may have been attributed to the fluctuating levels of homelessness in the past.

Year	V_S	U_S	$\hat{H}_S$
2034	19,294	4.988%	9,846
2044	24,447	4.874%	10,395
2074	24,153	4.530%	10,568

Table 4: Predicted Seattle Homeless Population $\hat{H}_S$ in 10, 20, and 50 years

Year	V_A	U_A	$\hat{H}_A$
2034	16,731	4.635%	1,403
2044	16,900	4.366%	1,362
2074	18,468	3.558%	1,234

Table 5: Predicted Albuquerque Homeless Population $\hat{H}_A$ in 10, 20, and 50 years

Because we wanted to minimize the cost, or error, with the few training data points that we had, we decided to iterate the model over 5,000 epochs, or steps, to reach a lower bound that would minimize the error. While a downside of increasing the number of steps is the greater cost of resources and run-time, we were able to achieve a mean-relative error of 0.177 for the Seattle homeless algorithm and an error of 0.0986 for the Albuquerque algorithm. In context, this means that, on average, our model’s predictions are off by 17% to actual values for predicting the Seattle homeless population. While these errors are somewhat high, the model is still reliable, especially considering the little data we had to train it to predict the homeless population 50 years into the future.

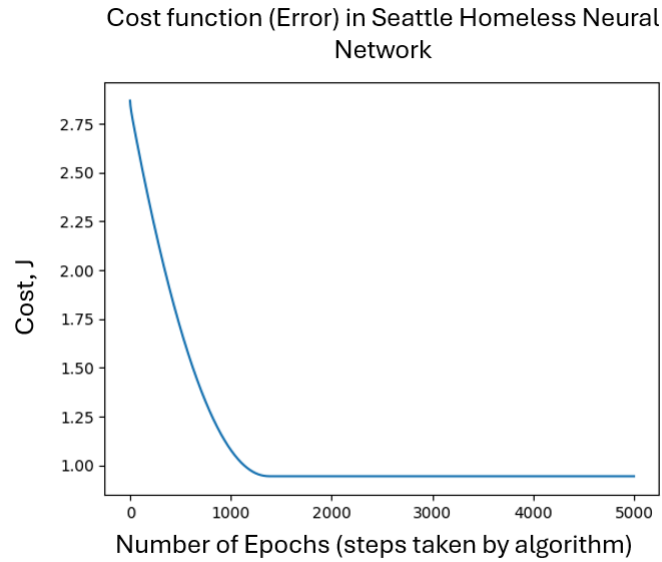


Figure 4: Graph of the cost function J in the Seattle Homeless Population Neural Network

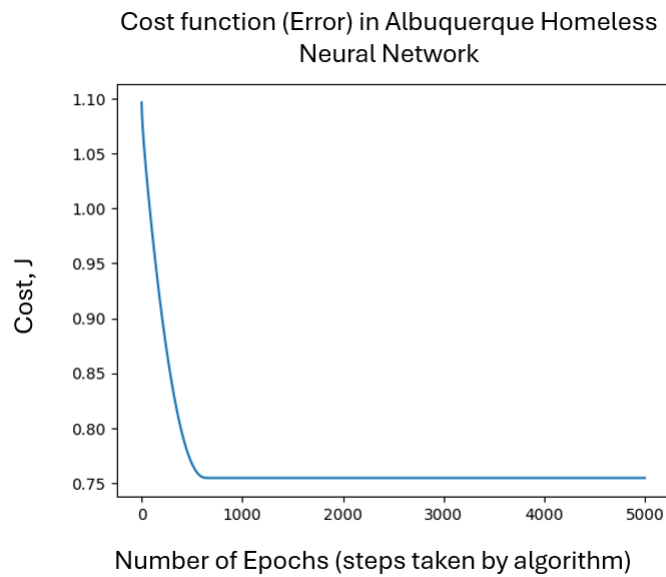


Figure 5: Graph of the cost function J in the Albuquerque Homeless Population Neural Network

Sensitivity Analysis

The part of our model that had the largest impact was the number of vacant housing units. To investigate the sensitivity, we increased the number of housing units by 1% in the Seattle model, which contributed to a decrease in the homeless population from 9,846 to 9,606, or a 2.4% decrease.

Strengths and Weaknesses

Sequential neural networks provide good predictions for multiple reasons. The main benefit of using a neural network was not having to choose stationary variables for time-series analysis. Considering variables involved in this project like unemployment, salary, and income are non-stationary, a time-series analysis will produce inaccurate results. Secondly, a neural network can make predictions using multiple variables, which allows it to take into consideration external factors and therefore be more accurate for long-term predictions. Finally, a sequential neural network, in comparison to traditional regression models, is able to extract relevant trends and patterns while disregarding skewed and incomplete data, which makes it more reliable for forecasting.

However, our machine learning model may be inaccurate due to the lack of training data that we were able to provide. While we had data values for vacant housing and historic unemployment, the lack of training data may have hindered our model from its full capabilities. Additionally, the lack of data points in our model's training set may have led to overfitting. Finally, the non-recursive nature of our model required us to estimate future values for unemployment. Compared to an auto-regressive time-series model, which recursively predicts future datapoints, a neural network requires an array of features in order to estimate a label.

Question 3: Rising from This Abyss

Restatement of the Problem

In this problem, we were tasked with developing a resilient long-term plan for a city to address homelessness, even in the face of external stresses such as natural disasters or economic recession.

Definitions

In order to accurately calculate the future unhoused population in our target regions, we must first establish the following definitions:

A person is **unhoused** if they do not have daily access to a permanent or rented housing unit. A person living in a lodging unit will be considered unhoused because lodging units are inherently un-permanent dwellings and do not contribute to the housing supply.

According to the Bureau of Labor Statistics, a person is classified as **unemployed** if they do not have a job, have actively looked for work in the prior 4 weeks, and are currently available for work (Bureau of Labor, n.d.).

Assumptions

1. *There will be no significant changes in policy regarding unhoused populations in the next 50 years.* Once again, we are unable to predict the actions of governments because they do not follow a definitive numerical trend.
2. *Future trends in unemployment rates, housing prices, and population will follow past trends.* Our models extrapolate future behavior based on past behavior and therefore require the continuation of previously exhibited behaviors to make robust predictions.
3. *There will be no humanitarian crisis happening in the next 50 years.* While events like COVID-19 can significantly affect unhoused populations, we are unable to predict whether they are going to happen or not.
4. *The simple random sample taken of monthly rent prices in a given metropolitan area accurately reflects the population of monthly rent prices.* We used Zillow's website to obtain a simple random sample (sample size = 30) of monthly rents in a city (Zillow, n.d.).
5. *Individuals who are homeless or on the verge of becoming homeless will first try to rent housing units before trying to buy them.* Buying a house is substantially more costly than renting a house. Since individuals strive to dwell in a housing unit, individuals who are homeless will first seek renting a housing unit before buying a housing unit.

Variables

Symbol	Definition	Unit
$\bar{r}$	Sampled average monthly rent in a given city	USD/month
S_r	Standard error of the sampled monthly rent in the given city	USD/month
h	Homelessness rate	%
t^*	t-score of the cutoff cost	n/a
C	Cutoff cost - the monthly rent of the rent percentile corresponding to the homelessness rate	USD/month
U	Monthly unemployment benefits in the given city	USD/month
U_r	Monthly unemployment benefits committable to monthly rent	USD/month

Table 6: Variable List for Question 3

Model Development

According to our Sequential Neural Network’s results from Q2, unemployment is expected to remain high for quite some time, despite our ARIMA’s predictions that vacant housing is also to remain generally high. From this, we deduced that the problem with housing, and the obstacle to their universal access, is not the availability of housing, but rather the pricing. According to endhomelessness.org, unattainable housing prices remain a leading cause of homelessness, confirming our findings and our hypothesis.

To address this problem, we developed a distribution approximation model to help cities understand how unemployment benefits policy helps, or fails to help, their homeless population work towards reattaining financial and housing stability. By using a simple random sample of rent pricing, our model constructs an approximation of the distribution of monthly housing rents in a given city. This distribution is assumed to be normal, so long as the sampling procedures abide by the restrictions of the Central Limit Theorem (CLT). The t-distribution is used to calculate t-scores, due to standard deviation being estimated by sample values. Based on the percentile of this distribution that corresponds to the homelessness rate in that city, our model determines if unemployment benefits are capable of assisting people experiencing homelessness into the lowest possible bracket of housing accommodation.

To find the t-score of the cutoff cost, which is the rent corresponding to the homeless percentage:

$$t^* = \text{inverse}T(h)$$

Next, the actual cutoff cost can be found using the sample standard error and sample mean:

$$C = S_r \cdot t^* + \bar{r}$$

Additionally, the average percent of monthly income committed to rent is 30% (Scott, 2023). Thus, the funds committable to rent from a given city's unemployment benefits can be modeled by:

$$U_r = 0.3 \cdot U$$

If the cutoff cost is above the average amount committable to rent, calculated by 30% of monthly unemployment benefit, then those benefits are deemed insufficient to providing for the city's population experiencing homelessness.

Results

To test our model, we took a simple random sample from the Zillow API's data on monthly rent rates in Seattle, Washington. Our sample consisted of 30 observations to satisfy the Central Limit Theorem (CLT)'s normality assumption prerequisite, allowing us to use 1-sample t-procedures on our observations as outlined in the Model Development section(Zillow, n.d.).

Variable	Value
$\bar{r}$	1690.66
S_r	321.24
h	0.8
t^*	-2.56
C	869.72
U	2684
U_r	805.2

Table 7: Results Table for Question 3

Because the monthly rent associated with the homelessness rate percentile C is greater than the monthly committable funds from Seattle unemployment benefits, our model deemed Seattle's unemployment benefits insufficient to sustaining housing for the Seattle homeless population. To make up for these pitfalls, we suggest that governments local to Seattle increase tax credits for reenters of properties within the rent percentile associated with the homeless population, which will likely lower monthly rent rates and bring more affordable housing to people experiencing homelessness under unemployment benefits.

Strengths and Weaknesses

One strength of our model is that the model is very adaptable to different situations, as it is based on the unemployment rate. If the unemployment rate changes drastically, the amount of money required for rent increases drastically, causing a direct change in the amount of gap that the government has to make up using its tax benefits. It is also adaptable to recessions and other economic downturns as we are using government intervention through tax decreases, which causes increases in aggregate demand and decreases in unemployment, and therefore homelessness. Another strength of our model is that the model is very simple as there are only a couple of independent variables.

We have a couple of weaknesses in our model. First, the simple random sample may have bias as data was selected using an online tool that could have eliminated houses that were renting but were not included as they did not have access to the internet. This would have caused a selection bias. Another weakness that this model has is that the model approximates a population distribution with a sampling distribution. This could cause inaccuracies in the true mean and standard deviation of the population which are two variables that we use to find our cut-off point.

Conclusion

First, we predicted housing availability in Seattle, Washington and Albuquerque, New Mexico in 2034, 2044, and 2074 using housing vacancy data. Within a Python environment, we analyzed historical data of housing vacancy using an ARIMA model. We determined the optimal term order for a high-confidence ARIMA, and conducted significance testing for coefficients and terms involved in the model. This model predicted that housing vacancy's fluctuations would subside in the coming decades, but that its typical value would remain constant and nonzero.

Continuing with our analysis of Seattle and Albuquerque, we next analyzed trends in homelessness using a sequential neural network, also in Python. Under the assumption of an unemployment cycle that reflects business cycles, we trained our neural network model to predict unemployment rates 10, 20, and 50 years from present-day. Finally, we assimilated data from the neural network output as well as housing vacancy results from our ARIMA model to estimate homelessness rates in 2034, 2044, and 2074. Our neural network model predicted marginal decreases in unemployment on both cities, which contributed to similarly marginal decreases in homelessness, despite generally sustained heights of housing vacancy predicted with our ARIMA model.

To address this discrepancy, we created an additional distribution-approximation model to assess whether these cities, specifically Seattle, Washington, is taking significant steps to resolve the homelessness crisis. Using samples of city rents, our model calculates the monthly funding required for an individual to inhabit a class of home corresponding to the homelessness rate in that city. If unemployment benefits of the city cannot reasonably cover this cost, then those benefits are deemed insufficient to aiding the homeless population in that

city. Our results indicate that Seattle is not providing sufficient benefits for their homeless to inhabit the lowest housing bracket in Seattle. We suggest to increase tax credits for renters of those homes if they decrease the rent rates of the homes they rent out, to provide people experiencing homelessness greater access to a sustainable cost of housed living.

By addressing the continuing crisis of homelessness, we may resolve as well the stagnant housing market which is, in many cities, starved for affordable options. The housing and homelessness crises are inherently intertwined, and will need to be resolved with a joint effort between local governments, real estate vendors, and consumers alike. If, through policy and practice, we can break these cycles, we may provide a home for once and for all.

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Code Appendix: Q1

```

1 import pandas as pd
2 import numpy as np
3 import matplotlib.pyplot as plt
4 from statsmodels.tsa.arima.model import ARIMA
5 from statsmodels.graphics.tsaplots import plot_acf, plot_pacf
6 from statsmodels.tsa.stattools import adfuller
7
8 # Read the data
9 data = pd.read_csv('Data\Seattle.csv', index_col=0, parse_dates=True)
10
11 # Plot the original time series
12 plt.figure(figsize=(10, 6))
13 plt.plot(data.index, data['Vacant units'], color='blue')
14 plt.title('Original Number of Vacant Units in Seattle')
15 plt.xlabel('Date')
16 plt.ylabel('Vacant units')
17 plt.show()
18
19 # Check for stationarity
20 result = adfuller(data['Vacant units'])
21 print('ADF Statistic:', result[0])
22 print('p-value:', result[1])
23 if result[1] > 0.05:
24     print('The data is non-stationary.')
25 else:
26     print('The data is stationary.')
27
28 # Plot ACF and PACF to determine the order of AR and MA terms
29 fig, ax = plt.subplots(2, 1, figsize=(10, 8))
30 plt.title('ACF and PACF of the Number of Vacant Units in Seattle')
31 plot_acf(data['Vacant units'], ax=ax[0])
32 plt.xlabel('Lag')
33 plt.ylabel('Auto-correlation')
34
35
36
37 plot_pacf(data['Vacant units'], ax=ax[1])
38 plt.xlabel('Lag')
39 plt.ylabel('Partial Auto-correlation')
40 plt.show()
41
42 # Fit ARIMA model
43 plt.figure(figsize=(10, 6))
44 model = ARIMA(data['Vacant units'], order=(2,0,1))
45 results = model.fit()
46 plt.title('Diagnostics of the ARIMA Model for the Number of Vacant Units in Seattle')
47 results.plot_diagnostics()
48
49
50 print(results.conf_int())
51 # Print model summary
52 print(results.summary())
53
54
55 # Forecasting
56 forecast_steps = 52 # Adjusted to match the length of the forecasted values
57 forecast = results.forecast(steps=forecast_steps, alpha=0.05)
58
59 # Print the forecasted values and confidence intervals
60 print("Forecasted values for the next", forecast_steps, "time steps:")
61 print(forecast)
62 print("Confidence intervals for the forecasted values:")
63
64 # Plotting forecast with confidence intervals
65 plt.figure(figsize=(10, 6))
66 plt.plot(data.index, data['Vacant units'], label='Original Data', color='blue')
67 plt.plot(forecast.index, forecast.values, label='Forecast', linestyle='dashed', color='red')
68 print(results)
69
70
71
72 plt.title('ARIMA Forecast for the Number of Vacant Units in Seattle')
73 plt.xlabel('Date')
74 plt.ylabel('Vacant units')
75 plt.legend()
76 plt.grid(True)
77 plt.show()
78

```

Code Appendix: Q2

```
1  import matplotlib.pyplot as plt
2  import numpy as np
3  import pandas as pd
4  import seaborn as sns
5  from sklearn.model_selection import train_test_split
6  from sklearn import preprocessing
7  from tensorflow.keras.models import Sequential
8  from tensorflow.keras.layers import Dense, Activation
9  from tensorflow.keras.losses import BinaryCrossentropy
10 from sklearn.metrics import classification_report
11
12 from google.colab import drive
13 drive.mount('/content/drive')
14
15 df = pd.read_csv('/content/drive/MyDrive/MB/SeattleHomeless.csv')
16
17 X = df[['Vacant Units','UE Rate']].values
18 Y = df['Homeless population'].values
19 Y = Y.reshape(-1,1)
20
21 scaler_X = preprocessing.StandardScaler()
22 scaler_X.fit(X)
23
24 scaler_Y = preprocessing.StandardScaler()
25 scaler_Y.fit(Y)
26
27 X_scaled = scaler_X.transform(X)
28 Y_scaled = scaler_Y.transform(Y)
29
30 # deletes model if it was created before
31 model = 0
32 # Create a neural network. Only the input layer
33 model = Sequential()
34 # Adds a layer with 1 node. This will be the output layer
35 model.add(Dense(1))
36 # Tells what error will be minimized
37 model.compile(loss='MSE')
38 # Trains the model, epochs is the number of steps the algorithmn takes (beyond
39 # the scope of our course, verbose - 0 is so that it does not print on the
40 # screen)
41 model.fit(X_scaled,Y_scaled,epochs=5000,verbose=0)
42 # creates a list with the value of the error after each step
43 J = model.history.history['loss']
44 # plots the error after each step
45 plt.plot(J)
46
47 Y_hat_scaled=model.predict(X_scaled)
48
49 Y hat = scaler_Y.inverse transform(Y hat scaled)
```

```
49 Y_hat = scaler_Y.inverse_transform(Y_hat_scaled)
50
51 print(X.shape)
52 print(Y.shape)
53 print(Y_hat.shape)
54
55 y = Y.reshape(-1)
56 y_hat = Y_hat.reshape(-1)
57
58 dy = pd.DataFrame()
59 dy['price'] = y
60 dy['predicted price'] = y_hat
61 dy.head()
62
63 all_errors = np.abs(y-y_hat)
64 sum_all_errors = np.sum(all_errors)
65
66 mean_absolute_error = sum_all_errors/len(all_errors)
67 print(mean_absolute_error)
68
69 # Can define the above things beforehand, but is not necessary
70
71 mean_absolute_error = np.sum(np.abs(y-y_hat))/y.shape[0]
72 print(mean_absolute_error)
73
74 scaler_Y
75 print(scaler_Y.mean_[0])
76 print(np.sum(y)/y.shape[0])
77
78 all_relative_errors=np.abs(y-y_hat)/y
79 # Ex: y = [3,4]   y_hat = [5,3]   np.abs(y-y_hat) = [2,1]
80 sum_all_relative_errors = np.sum(all_relative_errors)
81 mean_relative_error = sum_all_relative_errors/y.shape[0]
82 print(mean_relative_error)
83
84 mean_absolute_error/scaler_Y.mean_[0]
85
86 # this is the new set of examples. We only know the features. We have
87 # two examples in this case
88 X_new = np.array([[16731,4.988]])
89 # We scale the new features
90 X_new_scaled = scaler_X.transform(X_new)
91 # We compute the predicted scaled labels
92 Y_hat_new_scaled = model.predict(X_new_scaled)
93 # We unscale the predicted scaled labels
94 Y_hat_new = scaler_Y.inverse_transform(Y_hat_new_scaled)
95 # We reshape to one dimensions to plot
96 y_hat_new = Y_hat_new.reshape(-1)
```

```
97  print(y_hat_new)
98
99  # this is the new set of examples. We only know the features. We have
100 # two examples in this case
101 X_new = np.array([[24447,4.874]])
102 # We scale the new features
103 X_new_scaled = scaler_X.transform(X_new)
104 # We compute the predicted scaled labels
105 Y_hat_new_scaled = model.predict(X_new_scaled)
106 # We unscale the predicted scaled labels
107 Y_hat_new = scaler_Y.inverse_transform(Y_hat_new_scaled)
108 # We reshape to one dimensions to plot
109 y_hat_new = Y_hat_new.reshape(-1)
110 print(y_hat_new)
111
112 # this is the new set of examples. We only know the features. We have
113 # two examples in this case
114 X_new = np.array([[24153,4.53]])
115 # We scale the new features
116 X_new_scaled = scaler_X.transform(X_new)
117 # We compute the predicted scaled labels
118 Y_hat_new_scaled = model.predict(X_new_scaled)
119 # We unscale the predicted scaled labels
120 Y_hat_new = scaler_Y.inverse_transform(Y_hat_new_scaled)
121 # We reshape to one dimensions to plot
122 y_hat_new = Y_hat_new.reshape(-1)
123 print(y_hat_new)
```